Multiple IV Bolus Practice 2

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

The elimination half-life of an antibiotic is 3 hours and the apparent volume of distribution is 20% of the body weight. The therapeutic window for this drug is from 2 to 10 mcg/mL. Adverse toxicity is often observed at drug concentrations above 15 mcg/mL. The drug will be given by multiple IV bolus injections.

a Calculate the dose for an adult male patient (68 years old, 82 kg) with normal renal function to be given every 8 hours. (Hint, I started by assigning 10 mg/L as the maximum concentration at steady state).

b Calculate the maximum steady-state plasma concentration. (The regimen from part a: 138.16 mg every 8 hours to an 82-kg patient.)

c Calculate the minimum steady-state plasma concentration. (The regimen from part a gives a steady-state peak of 10 mg/L every 8 hours.)

d Calculate the average steady-state plasma concentration. (The regimen from part a: 138.16 mg every 8 hours to an 82-kg patient.)

e Does your regimen provide the desired peak and trough concentrations? If not, how would you adjust to achieve the desired concentrations? (The regimen from part a: 138.16 mg every 8 hours, giving a steady-state peak of 10 mg/L and a trough of 1.576 mg/L.)

A. Yes: both the peak and the trough stay below the 15 mcg/mL toxic level
B. No: the trough is below 2 mg/L; shorten the interval from 8 to 6 hours
C. No: the peak is too high; lower the dose and keep the 8-hour interval
D. No: the trough is too low; raise the dose and keep the 8-hour interval

e With the dosing interval decreased from 8 to 6 hours, what dose every 6 hours gives a steady-state peak of 10 mg/L? (82-kg patient, V_D 16.4 L.)

e On the new proposed regimen, 123 mg every 6 hours with an expected steady-state peak of 10 mg/L, what is the expected steady-state trough?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 138.16 mg

WHAT THE STEM GIVES

t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	20% of body weight	0.2 L per kg × 82 kg = 16.4 L
window	2 to 10 mcg/mL	the top, 10 mcg/mL, is the target C_{max}^{∞}
toxic above	15 mcg/mL	*not needed*
age	68 years old	*not needed: renal function is stated normal*
weight	82 kg	$V_D = 0.2 \times 82 = 16.4$ L
τ	every 8 hours	the exponent $k\tau = 0.231 \times 8$
C_{max}^{∞}	10 mg/L	the target peak, multiplied by V_D and $(1 - e^{-k\tau})$

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{3\ hr} = 0.231\ hr^{-1};\ V_D = (0.2)(82\ kg) = 16.4\ L$$

The rate constant is 0.693 over the half-life, and a volume given as 20% of body weight is 0.2 L per kilogram, so 0.2 × 82 kg = 16.4 L.

UNIT

$$C_{max}^{\infty} = 10\ mcg/mL = 10\ mg/L$$

The upper limit of the therapeutic window, 10 mcg/mL, is assigned as the steady-state peak; 1 mcg/mL is 1 mg/L, so it is 10 mg/L.

SETUP

$$C_{max}^{\infty} = \frac{D_0}{V_D(1 - e^{-k\tau})},\ \text{so}\ D_0 = C_{max}^{\infty}\ V_D\ (1 - e^{-k\tau})$$

The steady-state peak equation is rearranged for the dose: multiplying the target peak by the volume and by $(1 - e^{-k\tau})$ undoes the accumulation factor.

ALGEBRA

$$D_0 = (10\ mg/L)(16.4\ L)(1 - e^{-(0.231)(8)}) = (164\ mg)(0.8424) = 138.16\ mg$$

mg/L times L leaves mg, and $(1 - e^{-1.848}) = 0.8424$ is the fraction of each dose eliminated during one 8-hour interval; the dose is 138.16 mg.

DOES THE NUMBER MAKE SENSE?

A 10 mg/L peak in 16.4 L is 164 mg in the body; 8 hours is more than two half-lives of 3 hours, so over three quarters of each dose is gone by the next one, and the dose replaces that: a little under 164 mg, 138.16 mg.

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b

Her key

Answer: 10 mg/L

WHAT THE STEM GIVES

t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ \text{hr}^{-1}$
V_D	20% of body weight	0.2×82 kg = 16.4 L in the denominator
window	2 to 10 mcg/mL	*not needed for the number; the peak is compared with it afterwards*
toxic above	15 mcg/mL	*not needed*
D_0	138.16 mg	the numerator
τ	every 8 hours	the exponent $k\tau = 0.231 \times 8$
weight	82-kg	$V_D = 0.2 \times 82 = 16.4$ L

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max}^{\infty} = \frac{D_0}{V_D(1 - e^{-k\tau})}$$

The steady-state peak is the first-dose peak, D_0 over V_D, raised by the accumulation factor, 1 over $(1 - e^{-k\tau})$.

ALGEBRA

$$C_{max}^{\infty} = \frac{138.16\ \text{mg}}{(16.4\ \text{L})(1 - e^{-(0.231)(8)})} = \frac{138.16\ \text{mg}}{13.816\ \text{L}} = 10\ \text{mg/L}$$

mg over L gives mg/L; the dose was chosen in part a to give this peak, so the equation returns the assigned 10 mg/L, the top of the therapeutic window.

DOES THE NUMBER MAKE SENSE?

The first-dose peak $\frac{138.16}{16.4}$ is a little over 8 mg/L; 8 hours is more than two half-lives, so little accumulates and the steady-state peak is only a little higher: 10 mg/L, the top of the window the dose was chosen for.

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c

Her key

Answer: 1.576 mg/L

WHAT THE STEM GIVES

t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ \text{hr}^{-1}$
V_D	20% of body weight	*not needed: the peak is given, so no dose or volume enters*

window	2 to 10 mcg/mL	*not needed for the number; the trough is judged against the 2 mcg/mL floor afterwards*
toxic above	15 mcg/mL	*not needed*
C_{max}^{∞}	10 mg/L	the steady-state peak the trough is read down from
τ	every 8 hours	the exponent kτ = 0.231 × 8

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min}^{\infty} = C_{max}^{\infty}\, e^{-k\tau}$$

From the steady-state peak to the steady-state trough is one dosing interval of first-order decline, so the peak is multiplied by $e^{-k\tau}$.

ALGEBRA

$$C_{min}^{\infty} = (10\ \text{mg/L})\, e^{-(0.231)(8)} = (10\ \text{mg/L})(0.1576) = 1.576\ \text{mg/L}$$

The exponent $(0.231\ \text{hr}^{-1})(8\ \text{hr}) = 1.848$ is a pure number, so the trough keeps the unit of the peak; 8 hours is 2.67 half-lives and 15.8% of the peak remains.

> **DOES THE NUMBER MAKE SENSE?**
> Eight hours is more than two half-lives of 3 hours, so less than a quarter of the peak remains: 10 → 5 mg/L at three hours, half of that at six, and lower still at eight. 1.576 mg/L fits and falls below the 2 mcg/mL floor.

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d

Her key

Answer: 4.56 mg/L

WHAT THE STEM GIVES

t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ \text{hr}^{-1}$
V_D	20% of body weight	0.2 × 82 kg = 16.4 L in the denominator
window	2 to 10 mcg/mL	*not needed for the number*
toxic above	15 mcg/mL	*not needed*
D_0	138.16 mg	the numerator, with F = 1 for an IV dose
τ	every 8 hours	multiplies V_D and k in the denominator
weight	82-kg	$V_D = 0.2 \times 82 = 16.4$ L

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{V_D\, k\, \tau}, \text{ with } F = 1$$

The average over one interval at steady state is the dose absorbed per interval over the volume cleared per interval; an IV dose is wholly available, so F = 1.

ALGEBRA

$$C_{avg}{}^{\infty} = \frac{138.16\ \text{mg}}{(16.4\ \text{L})(0.231\ \text{hr}^{-1})(8\ \text{hr})} = \frac{138.16\ \text{mg}}{30.31\ \text{L}} = 4.56\ \text{mg/L}$$

L × hr^{-1} × hr leaves litres, so mg over L is mg/L; the average, 4.56 mg/L, lies between the trough, 1.576, and the peak, 10 mg/L, and below their midpoint because the decline is exponential.

> **DOES THE NUMBER MAKE SENSE?**
> The average lies below the 10 mg/L peak and, since the decline is exponential, below the midpoint of peak and trough; 138.16 mg in 8 hours over a clearance of 16.4 × 0.231 L/hr gives a little under 5 mg/L, so 4.56 fits.

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e

Her key

Answer: B. No: the trough is below 2 mg/L; shorten the interval from 8 to 6 hours

WHAT THE STEM GIVES

t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$, in $e^{-k\tau}$
V_D	20% of the body weight	*not needed: the peak and trough are given*
window	2 to 10 mcg/mL	the trough against 2, the peak against 10
toxic above	15 mcg/mL	*not needed: the window, not toxicity, is the test*
D_0	138.16 mg	*not needed: already inside the given peak and trough*
τ	every 8 hours	the exponent kτ = 0.231 × 8; the interval that is changed
$C_{max}{}^{\infty}$	10 mg/L	at the upper limit; the trough is 0.1576 of it
$C_{min}{}^{\infty}$	1.576 mg/L	below the 2 mg/L floor: the failing number

THE WORKING, ONE LINE AT A TIME

SETUP

$k = \frac{0.693}{3\ \text{hr}} = 0.231\ hr^{-1}$; window 2 to 10 mcg/mL = 2 to 10 mg/L; regimen:

$C_{max}{}^{\infty} = 10$ mg/L, $C_{min}{}^{\infty} = 1.576$ mg/L

mcg/mL and mg/L are the same size, so the window is 2 to 10 mg/L: the steady-state peak is judged against 10 and the trough against 2.

SETUP

Compare: peak 10 mg/L is at the upper limit, 10; trough 1.576 mg/L is below the lower limit, 2. The trough fails

The peak was assigned the top of the window, so it is inside. The trough is below 2 mg/L, so the antibiotic is sub-therapeutic at the end of each 8-hour interval.

ALGEBRA

Trough fraction at τ = 8 hr: $\frac{C_{min}^{\infty}}{C_{max}^{\infty}} = e^{-(0.231)(8)} = e^{-1.848} = 0.1576$; a trough of 2 mg/L would need a peak of $\frac{2 \text{ mg/L}}{0.1576} = 12.69$ mg/L, above 10

With the interval fixed, the trough is always 0.1576 of the peak, whatever the dose, so lifting the trough to 2 mg/L lifts the peak to 12.69 mg/L, outside the window.

ALGEBRA

$\frac{10 \text{ mg/L}}{2 \text{ mg/L}} = 5$; time for the peak to fall to the floor: $t = \frac{\ln 5}{k} = \frac{1.609}{0.231 \text{ hr}^{-1}} = 6.97$ hr, so τ must be under 7 hr; try τ = 6 hr

Starting at the 10 mg/L peak, the level reaches the 2 mg/L floor after 6.97 hours, so an interval under that keeps the trough inside. Six hours is the practical interval just under it.

ALGEBRA

C_{min}^{∞} at τ = 6 hr: (10 mg/L) $e^{-(0.231)(6)}$ = (10 mg/L) $e^{-1.386}$ = (10 mg/L)(0.25) = 2.5 mg/L, inside 2 to 10

Six hours is two half-lives of 3 hours, so a quarter of the peak remains at the trough: 2.5 mg/L, above the 2 mg/L floor.

SETUP

Decision: no, the 8-hour trough of 1.576 mg/L is below 2 mg/L; shorten the interval from 8 to 6 hours, which gives a trough of 2.5 mg/L

The trough-to-peak ratio is set by the interval alone, so the correction is a shorter interval, not a larger dose.

> **DOES THE NUMBER MAKE SENSE?**
> The 15 mcg/mL toxic level is not the test; the window is. The dose scales peak and trough together, so at 8 hours a trough of 2 needs a peak of 12.69, over 10; only a shorter interval moves the trough up alone, and 6 hours gives 2.5 mg/L.

The trough, 1.6 mg/L, is below the 2 to 10 mcg/mL window, so the drug is sub-therapeutic for part of each interval. A 6-hour interval gives a trough of 2.5 mg/L.

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e

Her key

Answer: 123 mg

WHAT THE STEM GIVES

t½	3 hours	gives $k = \frac{0.693}{3} = 0.231 \text{ hr}^{-1}$
V_D	20% of body weight	given as 16.4 L for the 82-kg patient
window	2 to 10 mcg/mL	the top, 10 mcg/mL, is the target peak
toxic above	15 mcg/mL	*not needed*
τ	every 6 hours	the exponent kτ = 0.231 × 6
C_{max}^{∞}	10 mg/L	the target peak, multiplied by V_D and $(1 - e^{-k\tau})$

weight	82-kg	*not needed: V_D is already given as 16.4 L*
V_D	16.4 L	multiplies the target peak

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_0 = C_{max}^{\infty}\, V_D\,(1 - e^{-k\tau}),\ \text{with } \tau = 6 \text{ hr}$$

The same rearranged peak equation as for the 8-hour regimen; only the interval changes, and a shorter interval leaves a smaller $(1 - e^{-k\tau})$, so a smaller dose holds the same peak.

ALGEBRA

$$D_0 = (10 \text{ mg/L})(16.4 \text{ L})(1 - e^{-(0.231)(6)}) = (164 \text{ mg})(0.75) = 123 \text{ mg}$$

mg/L times L leaves mg; 6 hours is two half-lives, so three quarters of each dose is eliminated during an interval and the dose that holds a 10 mg/L peak is 123 mg.

ALGEBRA

$$\text{Check: } C_{max}^{\infty} = \frac{123 \text{ mg}}{(16.4 \text{ L})(1 - e^{-(0.231)(6)})} = \frac{123 \text{ mg}}{12.3 \text{ L}} = 10 \text{ mg/L}$$

Putting the dose back into the peak equation returns the assigned 10 mg/L, so the new regimen keeps the peak at the top of the window.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is two half-lives of 3 hours, so three quarters of each dose is gone before the next; the dose replaces three quarters of the 164 mg a 10 mg/L peak puts in 16.4 L: 123 mg, smaller than the 8-hour dose.

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e

Her key

Answer: 2.5 mg/L

WHAT THE STEM GIVES

t½	3 hours	gives $k = \frac{0.693}{3} = 0.231 \text{ hr}^{-1}$
V_D	20% of body weight	*not needed: the peak is given, so no dose or volume enters*
window	2 to 10 mcg/mL	*not needed for the number; the trough is judged against the 2 mcg/mL floor afterwards*
toxic above	15 mcg/mL	*not needed*
D_0	123 mg	*not needed: already inside the given peak*
τ	every 6 hours	the exponent $k\tau = 0.231 \times 6$
C_{max}^{∞}	10 mg/L	the steady-state peak the trough is read down from

THE WORKING, ONE LINE AT A TIME

SETUP

$C_{min}^{\infty} = C_{max}^{\infty}\, e^{-k\tau}$, with $\tau = 6$ hr

From the steady-state peak to the steady-state trough is one 6-hour interval of first-order decline, so the peak is multiplied by $e^{-k\tau}$.

ALGEBRA

$C_{min}^{\infty} = (10 \text{ mg/L})\, e^{-(0.231)(6)} = (10 \text{ mg/L})(0.25) = 2.5 \text{ mg/L}$

Six hours is two half-lives, so a quarter of the peak remains; 2.5 mg/L is above the 2 mcg/mL lower limit, so the new regimen of 123 mg every 6 hours sits inside the window at both ends.

DOES THE NUMBER MAKE SENSE?

Six hours is two half-lives of 3 hours, so the peak halves twice: 10 → 5 → 2.5 mg/L. The regimen is meant to sit inside the window of 2 to 10 mcg/mL, so a trough above the 10 mg/L peak or below 2 mg/L has the wrong line.